A Robust and Interpretable Deep Learning Framework for Crop Health Assessment Using Hybrid CNN Feature Stacking

Comments

1. The manuscript presents a hybrid CNN feature stacking approach using ResNet50, EfficientNetB0, and MobileNetV2. While technically sound, the idea of combining multiple pre-trained CNN backbones is not new and has been explored in prior works on ensemble learning and feature fusion.
2. The use of 3 models will increase the complexity of the proposed system and how is the output of the 3 modules features integrated and given to the next section. From which layer is the features extracted in each module.
3. The framework essentially repurposes existing CNN architectures and standard dimensionality reduction techniques (PCA, Logistic Regression). There is limited methodological innovation beyond stacking features.
4. The use of transfer learning techniques are conventional and the training details and the hyperparameter settings for training to be elaborated with respect to optimizer and the learning rate.
5. The paper is more theoretical and the implementation results to be included.
6. The detailed architecture diagram of (MLP-Full, MLP-PCA, PCA-Logistic to be included and elaborated.
7. The figure depicting the result
    1. Sample images of dataset
    2. Preprocessed output
    3. Training curve
    4. Feature extraction result
    5. Confusion matrix
8. Accuracy, precision, recall, F1, and AUC-ROC are not reported. Ensure that comparative tables with baseline models are included.
9. Visualization of Predicted features are to included
10. For publication, the authors should either (a) extend the work to field datasets to demonstrate real-world robustness, or (b) introduce methodological innovations beyond straightforward feature stacking.

11. The manuscript is well written and technically correct, but it lacks strong novelty and significant improvement over existing methods.